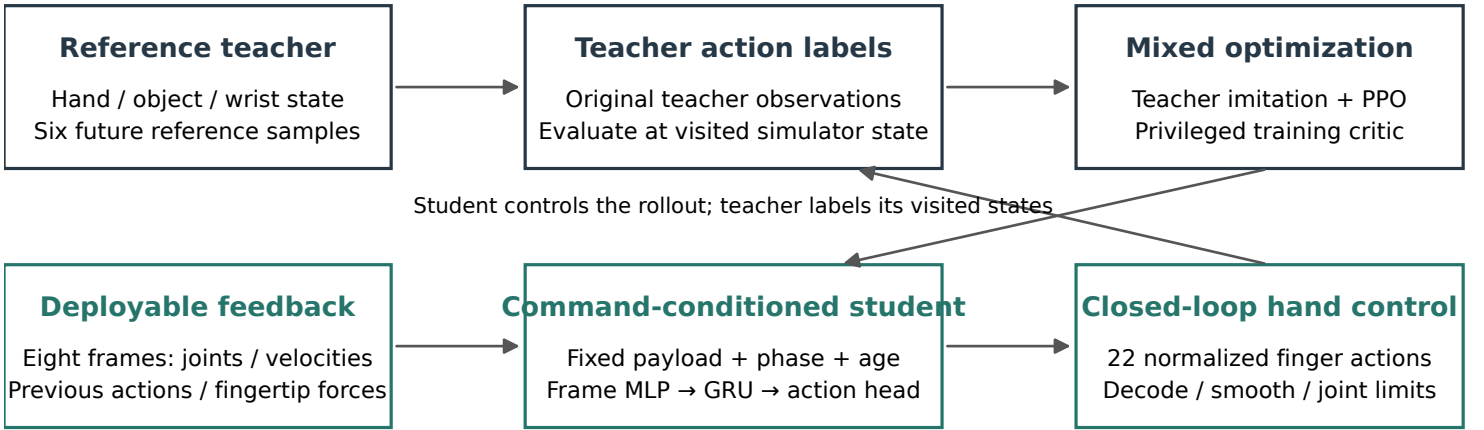
Reference teacher
Hand / object / wrist state
Six future reference samples
Teacher action labels
Original teacher observations
Evaluate at visited simulator state
Mixed optimization
Teacher imitation + PPO
Privileged training critic
Student controls the rollout; teacher labels its visited states
Deployable feedback
Eight frames: joints / velocities
Previous actions / fingertip forces
Command-conditioned student
Fixed payload + phase + age
Frame MLP → GRU → action head
Closed-loop hand control
22 normalized finger actions
Decode / smooth / joint limits